

A Stochastic Dual-Timescale Model of Learning: From Motor Neurons to Language Models

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I. Introduction

This project sits at the intersection of two areas we find compelling: the mathematical theory of stochastic differential equations and the question of why modern AI systems fail in predictable ways such as catastrophic forgetting. Understanding these failures requires tools that go beyond deterministic analysis, which is what motivated this project for this class.

Learning does not happen at a single speed. A student who crams the night before an exam learns quickly and forgets within a week. One who studies in small chunks over weeks retains the material for months. This contrast, fast acquisition versus slow retention, is not unique to human memory. It appears in motor adaptation, in gradient-based optimization, and in the training dynamics of artificial neural networks. Fast learning and slow learning are happening simultaneously, and understanding how they interact, especially when there is noise and randomness involved, is what this project is about.

Smith, Ghazizadeh, and Shadmehr (2006) gave this idea a mathematical foundation through a two-rate model of motor adaptation. Their model explains two observations no single-rate model can account for. *Savings*: a person relearns a skill faster the second time, even after a rest period, meaning something was retained despite a drop in performance. *Spontaneous recovery*: an earlier learned behavior partially reappears after being overwritten, meaning the first memory was suppressed, not erased. Both require at least two memory traces at different timescales.

Biologically, the fast component corresponds to the cerebellum, rapid adaptation but easily disrupted and the slow component to the motor cortex and striatum, gradual but durable. This maps onto broader Complementary Learning Systems theory [?], which describes the hippocampus as a fast temporary store and the neocortex as a slow permanent memory.

This project extends the Smith et al. model in two directions. First, we reformulate in continuous time and introduce rest periods by making the learning target time-varying. Second, we append Wiener process noise to each component and analyze the resulting system using Ito calculus and the Lyapunov equation, deriving three analytical results that connect to how noise behaves in artificial neural networks and language models.

II. Methodology

A. Continuous-Time Reformulation

The original Smith et al. (2006) model updates trial by trial in discrete time. We reformulate it as a continuous-time ODE system to access tools like eigenvalue analysis, phase planes, and Ito's calculus:

$$\frac{df}{dt} = -a_f f + b_f (T(t) - f - s), \quad (1)$$

$$\frac{ds}{dt} = -a_s s + b_s (T(t) - f - s), \quad (2)$$

where $f(t)$ and $s(t)$ are the fast and slow adaptation levels, $T(t)$ is the learning target, and $T-f-s$ is the shared error signal. The constraints $a_f > a_s$ and $b_f > b_s$ must always hold. If either is violated, the model collapses to a single time scale and cannot reproduce savings or differential forgetting.

In vector form $d\mathbf{X}/dt = J\mathbf{X} + \mathbf{c}(t)$, where J is the Jacobian. With parameters $a_f = 0.9$, $a_s = 0.1$, $b_f = 0.8$, $b_s = 0.2$, the system has a unique globally stable equilibrium at $f^* \approx 0.229$, $s^* \approx 0.514$, confirmed by $\text{tr}(J) < 0$ and $\det(J) > 0$. The eigenvalues $\lambda_1 \approx -0.194$ and $\lambda_2 \approx -1.806$ encode the two timescales with ratio ≈ 9.3 .

B. Rest Period Extension

The basic model assumes learning never stops. To make it realistic, we replace the fixed target with a switching signal:

$$T(t) = \begin{cases} 1 & \text{learning period} \\ 0 & \text{rest period} \end{cases}$$

The Jacobian is identical in both phases so the eigenvalues do not change. During rest, both components decay toward zero, but at very different rates: the fast component decays nine times faster than the slow one because $|\lambda_2| \approx 9|\lambda_1|$. This produces savings and differential forgetting without adding any new parameters – both emerge purely from the eigenvalue structure.

C. Stochastic Extension

Real learning systems are noisy. Ion channels in neurons fire randomly, SGD computes gradients on random mini-batches, and LLM temperature controls randomness in output sampling. All three are continuous, mean-zero, accumulating perturbations – exactly what a Wiener process models. We append independent Wiener processes to each equation:

$$df = [-a_f f + b_f(T - f - s)] dt + \sigma dW_f, \quad (3)$$

$$ds = [-a_s s + b_s(T - f - s)] dt + \sigma dW_s. \quad (4)$$

The drift is unchanged; noise is purely additive. This gives three analytical results.

Proposition 1 (Mean preservation): Taking expectations and using $\mathbb{E}[dW] = 0$, the noise term vanishes and the mean satisfies the same ODE as the deterministic system. Noise does not bias the learned outcome.

Proposition 2 (σ^2 scaling): The covariance $P(t)$ satisfies the Lyapunov equation $dP/dt = JP + PJ^\top + \sigma^2 I$. At stationarity, writing $P_\infty = \sigma^2 Q$ and substituting gives $JQ + QJ^\top + I = 0$, which contains no σ . Therefore stationary variance scales exactly as σ^2 , with $\text{Var}_\infty(f) = \sigma^2 Q_{11} \approx 0.771\sigma^2$ and $\text{Var}_\infty(s) = \sigma^2 Q_{22} \approx 2.343\sigma^2$.

Proposition 3 (Slow has higher variance): Since $Q_{22} > Q_{11}$, the slow component accumulates more stationary

variance than the fast one, ratio ≈ 3.04 . The intuition: a weaker restoring force means noise kicks linger longer before decaying. Fast forgets noise quickly; slow accumulates it.

D. Dataset Selection

In order to ensure training results are readable to extract insights, we choose to use a more simplistic dataset for training purposes to more easily see how tuning certain neural network parameters and hyperparameters will affect the stochastic dual learning training outcomes. For this purpose, we choose to use HuggingFace's MNIST(Modified National Institute of Standards and Technology) dataset.

The dataset consists of 70,000 examples of grayscale images depicting handwritten digits from 0 to 9. The images are normalized to fit a 28 by 28 pixel grid for memory and model training purposes.



Fig. 1 MNIST Dataset Sample

E. Neural Network Training

A big part of neural network training is to fine tune parameters to ensure that the Loss is minimized. And in our case of stochastic equation testing, ensuring that noise doesn't overwhelm the results. Loss in a neural network is a numerical value measuring the difference between predicted output and actual target values, essentially quantifying how "wrong" the model is. It serves as a cost function during training that is fed back into the loop as part of the feedback.

Our goal was to explore how the parameters should be adjusted, and how well the slow and fast components of learning would work together to ensure loss still behave normally with the benefits of the added noise. The 3 main parameters we tested various levels for were the learning rate, diffusion coefficient (σ), and *as_val*.

Most notably we tested 3 vastly different σ values: 0.02, 0.002, 0.001. At the 0.02 rate, Most notably at the 0.02 and 0.002 values, the noise significantly overpowered and ordered trends within the loss graph, with 0.02 rendering the loss graph almost entirely random, While at the 0.002 value, after the first third of training steps, the noise completely overtook any learning signals. Similarly to the diffusion coefficient, we had to increase the learning rates slightly to ensure that the signal was more powerful in the stochastic training iterations, to overpower the effect of noise snowballing

at each iteration.

Something discussed earlier is the difficulty in balancing the slow learning(s) and fast learning(f) components, especially in terms of the "memory" retention in the slow learning components to better emulate biological learning behavior. As slow learning naturally has a much slower decay rate in information retention, we had to decrease our initial as_val (decay rate) of just the slow learning function to allow for better information retention through each training step of the neural network.

III. Results

A. Deterministic Model

Figure 3 shows the time series and phase plane. Starting from zero, the fast component rises steeply and settles near $f^* \approx 0.23$, while the slow component rises gradually and dominates at $s^* \approx 0.51$. All five trajectories from different initial conditions converge to the same equilibrium, confirming global stability.

B. Rest Period Extension

Figure 4 shows uniform and non-uniform rest schedules. In both cases the fast component nearly resets during each rest period while the slow component retains most of its value. The rising baseline of $x(t)$ across sessions confirms savings. The behavior holds even with irregular session lengths, showing robustness of the eigenvalue-driven mechanism.

C. Stochastic Extension

Mean preservation. Figure 6 shows Euler-Maruyama simulation at $\sigma \in \{0.05, 0.15, 0.30\}$ with $n = 150$ paths each. At every noise level the mean of the paths overlaps the deterministic solution, confirming Proposition 1. Individual paths spread more as σ increases but the average does not drift.

σ^2 *scaling law.* Figure 5 (left) shows stationary variance against σ on a log-log scale, estimated from $n = 300$ paths run to $t = 150$. The log-log slopes are approximately 1.98 and 2.02 for the fast and slow components respectively, confirming Proposition 2. The Lyapunov analytical prediction (dashed lines) matches the simulation closely across all noise levels.

Slow has higher variance. Figure 5 (right) plots $\text{Var}_\infty(f)$ and $\text{Var}_\infty(s)$ on the same axes. The slow component consistently exceeds the fast at every σ , with ratio ≈ 3.04 , matching the analytical Q_{22}/Q_{11} and confirming Proposition 3.

D. Neural Network Loss & Stochastic Learning Advantages

Shown in the figure below is a comparison of the loss curves and parameter changes between stochastic(in red) and deterministic(in blue) training behaviors. We can see that as expected, the stochastic training is much more volatile,

with varied changes to both loss and parameter through the hundreds of training steps. While stochastic loss still generally conforms to loss curve of the deterministic training, having such volatility can actually mean good things in certain circumstances. It can lead to better generality of a model, especially when provided small training sets, and having limited computer resources to train a model. You can think of the process as marble rolling down a ramp. In a deterministic process, it's likely the marble stops at the first minimum of loss, but in a stochastic process you're shaking the marble giving it a higher chance of finding an even lower global minimum.

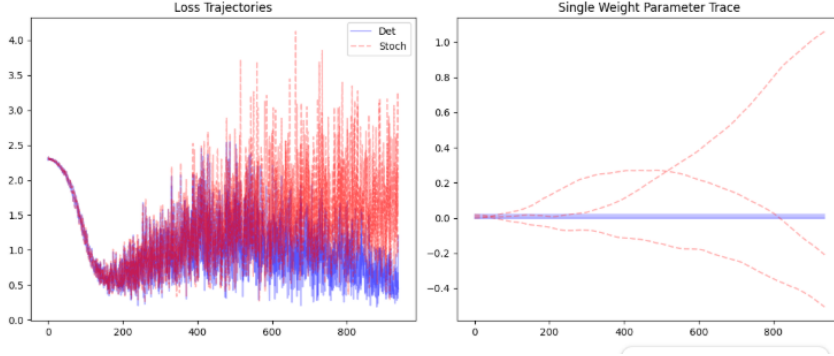


Fig. 2 Neural Network Loss and Parameter Changes Deterministic vs. Stochastic Learning

IV. Conclusion & Future Work

This project developed and analyzed a continuous-time stochastic dual-timescale model of learning, motivated by the neuroscience of motor adaptation and connected to current questions in AI.

The deterministic analysis established a unique globally stable equilibrium with two distinct timescales encoded in the eigenvalues, ratio ≈ 9.3 . The rest period extension reproduced savings and differential forgetting without any new parameters, both emerge purely from the eigenvalue structure. The stochastic extension derived three analytical results: mean preservation, σ^2 scaling of stationary variance, and the counterintuitive finding that the slow component accumulates more noise than the fast one. All three were verified numerically using Euler-Maruyama simulation. The neural network experiment confirmed these predictions in a real training setting, stochastic training showed higher volatility but the mean loss tracked the deterministic solution, consistent with Proposition 1, and the weight parameter spread across runs matched the higher slow-component variance predicted by Proposition 3.

A key insight from both the analytical model and the experiment is that noise does not affect both components equally. The component responsible for long-term retention, the slow one, is also the most affected by noise. This means in any noisy learning system, long-term outcomes are inherently less reproducible than short-term ones.

The standard alternative to SDE analysis is purely numerical simulation. This approach can reveal the σ^2 scaling empirically but has key limitations. It cannot prove the exact scaling exponent analytically, it cannot explain *why* the slow component has higher variance since that connection requires the Lyapunov equation, and it cannot generate

predictions for untested systems. The SDE framework converts empirical observations into theorems and produces falsifiable predictions, such as late transformer layers being noisier than early ones, that arise purely from the analytical structure.

Several directions remain open. First, extending to state-dependent noise $\sigma(\mathbf{X})$ would better capture heavier-tailed output distributions observed in real LLMs. Second, a three-component model with a separate replay mechanism, where the slow component receives processed summaries rather than raw error signals, mirroring how the neocortex receives input from the hippocampus during sleep, would more faithfully capture the biological separation that current AI architectures lack. Third, the prediction that late transformer layers show higher variance across runs than early ones is directly testable with any open-weight model and would connect this framework to AI interpretability research.

V. Appendix

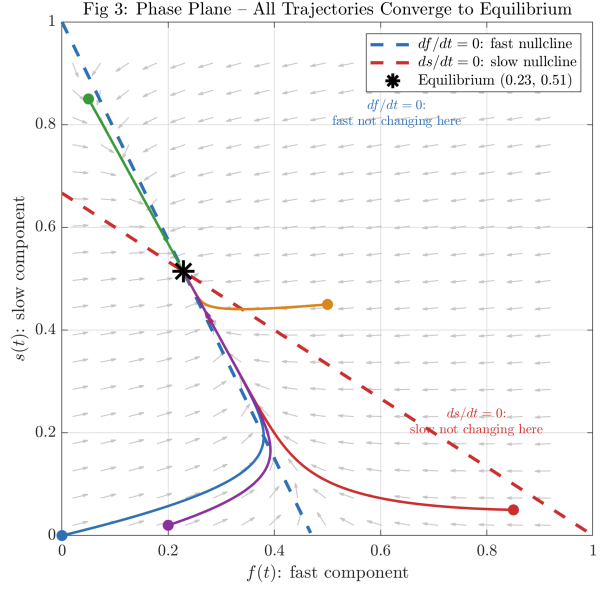
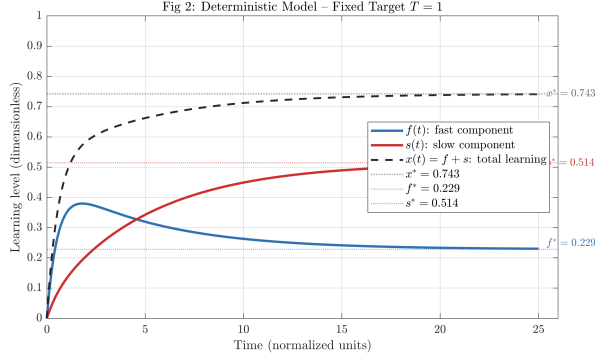


Fig. 3 Left: deterministic time series. Fast (blue) settles at $f^* \approx 0.23$; slow (red) dominates at $s^* \approx 0.51$. Right: phase plane – all trajectories converge to the equilibrium (star).

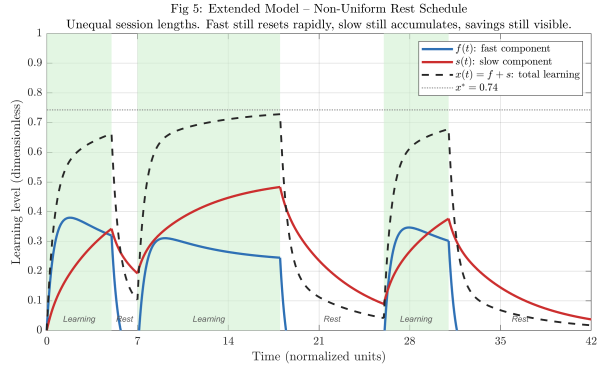
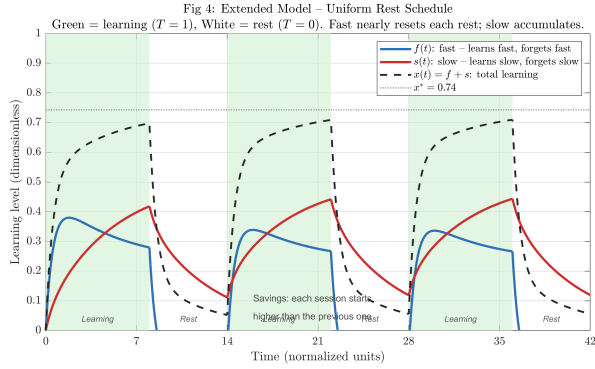


Fig. 4 Rest period simulations. Left: uniform schedule. Right: non-uniform schedule. Fast (blue) nearly resets during rest; slow (red) accumulates. Rising baseline confirms savings.

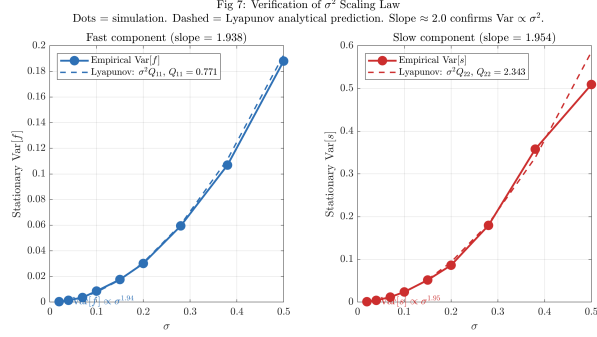


Fig. 5 Left: log-log σ^2 scaling, slopes ≈ 2.0 . Right: slow variance (red) exceeds fast (blue) at every σ , ratio ≈ 3.04 .

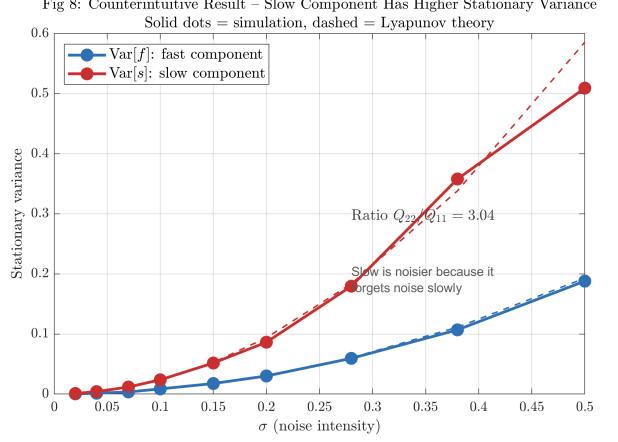


Fig. 6 SDE Simulation at Three Noise Levels
Mean (black) tracks deterministic (red dashed) at every σ . Paths (light blue) spread more as σ grows.

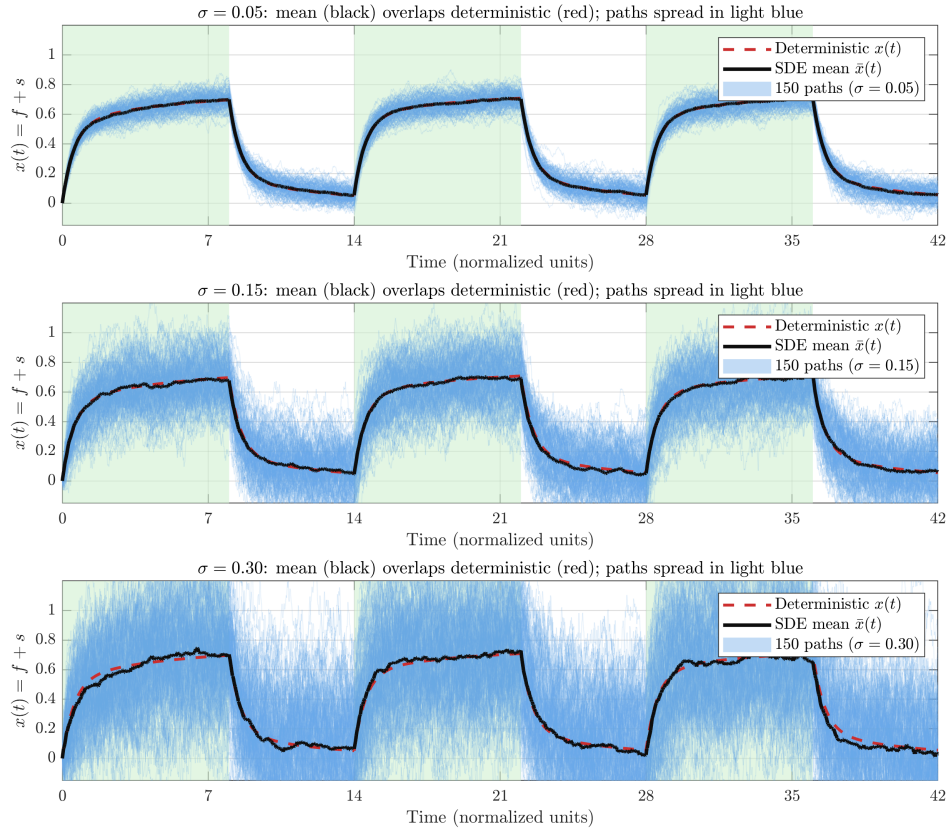


Fig. 6 Euler-Maruyama simulation at $\sigma \in \{0.05, 0.15, 0.30\}$, $n = 150$ paths. Mean (black) overlaps deterministic (red dashed) at every noise level.

An interactive application allowing real-time exploration of the model is available at:

<https://hima03.shinyapps.io/app2/>

The app has three tabs. The first shows the deterministic model with rest periods, sliders for all four parameters update the time series, equilibrium values, and eigenvalues live, with a warning if the biological constraints are violated. The second shows SDE sample paths with adjustable σ and number of paths, where the mean is overlaid against the deterministic solution to visually confirm mean preservation. The third computes the σ^2 scaling law and plots stationary variance on a log-log scale with the fitted slope.

VI. References

- Smith, M. A., Ghazizadeh, A., & Shadmehr, R. (2006). Interacting adaptive processes with different timescales underlie short-term motor learning. *PLoS Biology*, 4(6), e179.
- McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. (1995). Why there are complementary learning systems in the hippocampus and neocortex. *Psychological Review*, 102(3), 419–457.